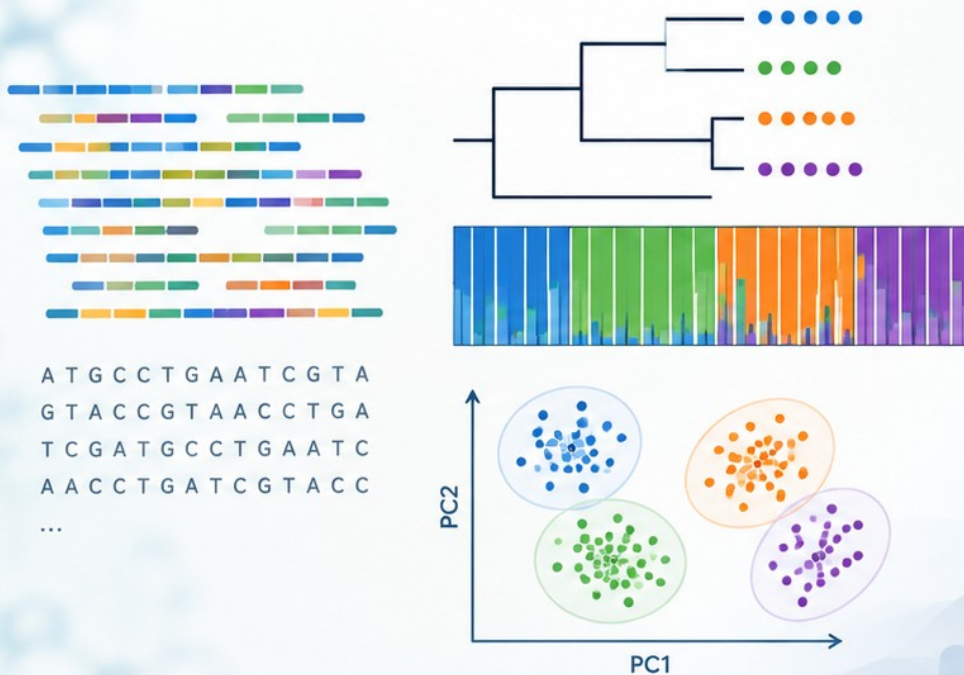


Population differentiation via F_{st} and F-statistics

PR Stats Course

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia



Nucleotide diversity and heterozygosity

$$\pi = \frac{\sum_{i=1}^S 2 \cdot p_i \cdot q_i}{L}$$

Where:

- S = Number of segregating sites (variable positions) in the population.
- p_i = Frequency of the reference allele at site i .
- q_i = Frequency of the alternate allele at site i ($1 - p_i$).
- L = Total number of callable (non-missing) sites in the genome.
- The $2 \cdot p \cdot q$ term is the probability that two randomly drawn chromosomes differ at that site.

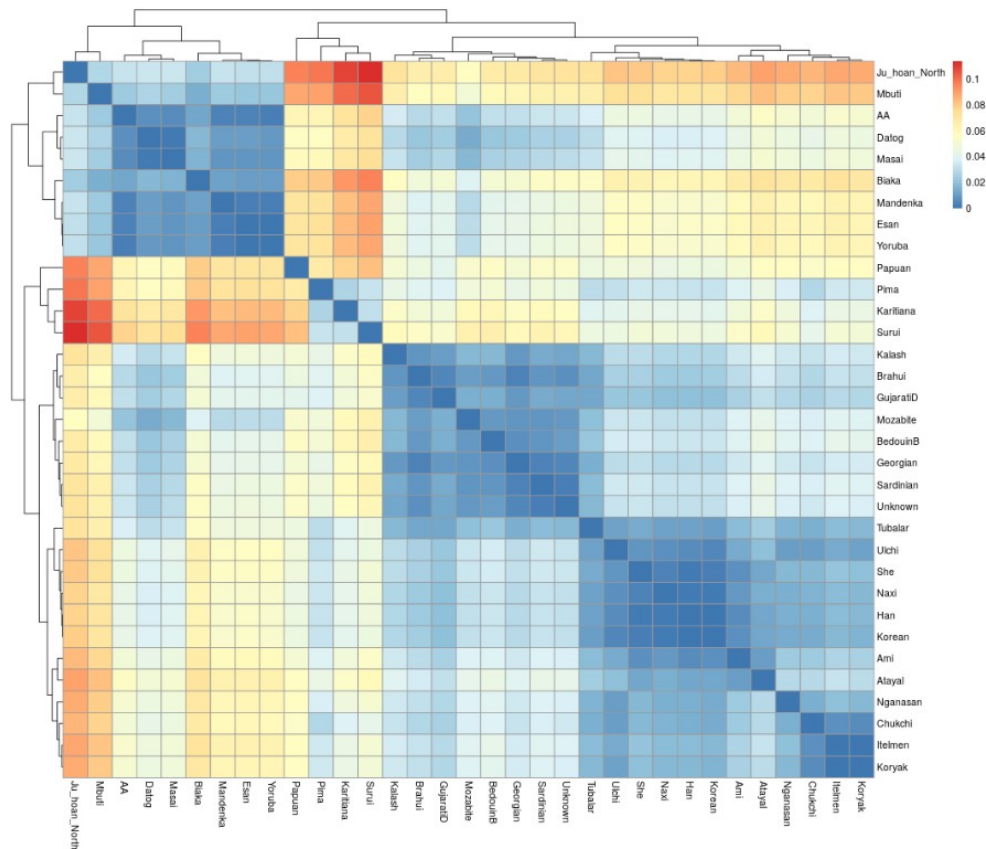
Importantly, one should not mix up individual heterozygosity and population heterozygosity

Fixation index F_{st}

- F_{st} is a measure of population differentiation due to population structure
- Expressed through nucleotide diversity π as relative nucleotide diversity between populations minus average nucleotide diversity within populations. If we have two populations 1 and 2, then:

$$F_{st} = \frac{\pi_{12} - \frac{\pi_1 + \pi_2}{2}}{\pi_{12}} \sim \frac{\sigma_{subpops}^2}{\sigma_{total}^2}$$

```
1 library("admixtools")
2 library("tidyverse")
3
4 admixtools::extract_f2("AADR", outdir = "f2_AADR")
5
6 f2_aadr <- read_f2("f2_AADR")
7 fst_aadr <- fst(f2_aadr)
8 fst_aadr
9
10 mat <- f2(f2_aadr, unique_only = F) %>%
11   select(-se) %>%
12   pivot_wider(names_from = pop2, values_from = est) %>%
13   column_to_rownames("pop1") %>%
14   as.matrix()
15
16 library("pheatmap")
17 pheatmap(mat, cluster_rows = TRUE, cluster_cols = TRUE)
```



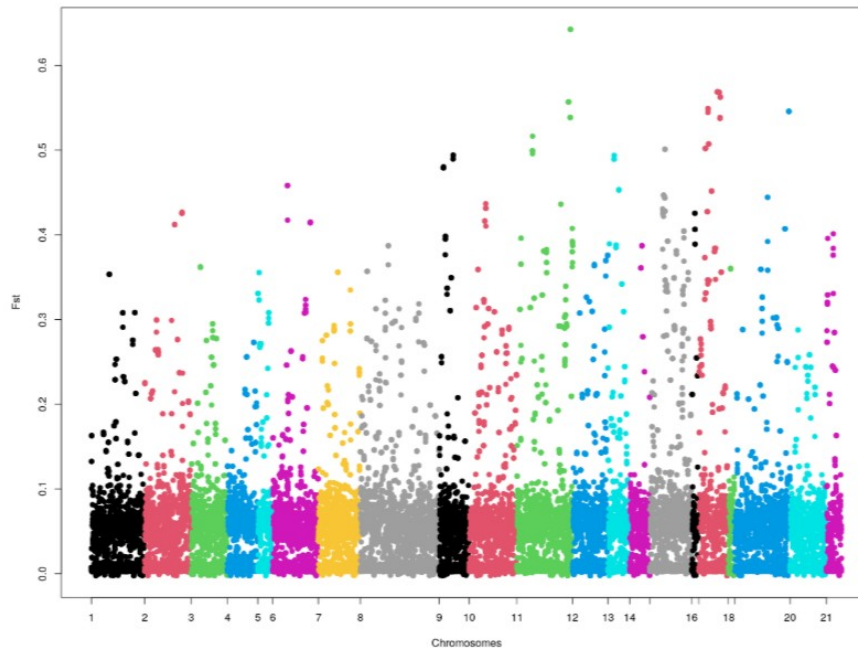
Example of Fst computation from 1000G project

We will use ANGSD (Korneliussen T, 2014) and compute Fst between CEU and YRI populations from the 1000G project.

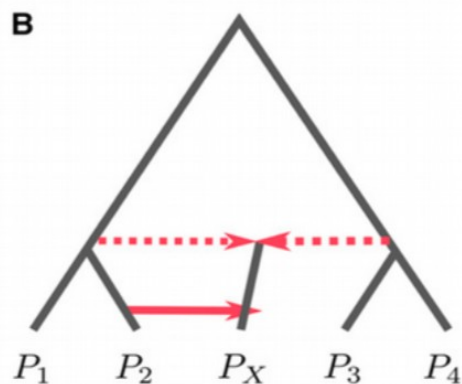
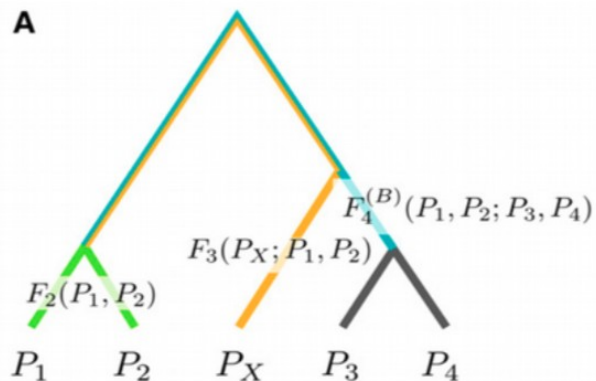
```
1 angsd -b 1000G_CEU_bam_list.txt -anc anc.fa -out CEU -dosaf 1 -gl 1
2 angsd -b 1000G_YRI_bam_list.txt -anc anc.fa -out YRI -dosaf 1 -gl 1
3 realSFS CEU.saf.idx YRI.saf.idx > CEU.YRI.ml
4 realSFS fst index CEU.saf.idx YRI.saf.idx -sfs CEU.YRI.ml -fstout CEU_YRI
5 realSFS fst stats CEU_YRI.fst.idx
6 realSFS fst stats2 CEU_YRI.fst.idx -win 50 -step 10 > slidingwindow_win50_step10
```

and finally we will plot the results using custom R scripts:

```
1 df <- read.delim("slidingwindow_win50_step10", header = FALSE)
2 df <- df[-1, ]
3 df$V2 <- as.numeric(df$V2)
4 df$V3 <- as.numeric(df$V3)
5 df$V5 <- as.numeric(df$V5)
6 df <- df[order(df$V2, df$V3), ]
7 rownames(df) <- seq(1, dim(df)[1], 1)
8 plot(df$V5, col = df$V2, xlab = "Chromosomes", ylab = "Fst", xaxt = "n")
9 myticks <- as.numeric(rownames(df[!duplicated(df$V2), ]))
10 axis(side = 1, at = myticks, labels = seq(1, 21, 1))
```



F-statistics: underlying assumptions



- Treeness test: are populations related in a tree-like fashion (Reich et al. 2009)?
- Admixture test: is a particular population descended from multiple ancestral populations (Reich et al. 2009)?
- Admixture proportions: what are the contributions from different populations to a focal population (Green et al. 2010; Haak et al. 2015)?
- Number of founders: how many founder populations are there for a certain region (Reich et al. 2012; Lazaridis et al. 2014)?
- Complex demography: how can mixtures and splits of population explain demography (Patterson et al. 2012; Lipson et al. 2013)?
- Closest relative: what is the closest relative to a contemporary or ancient population (Raghavan et al. 2014)?

(Peter, 2016)

F2-statistic and F3-statistic

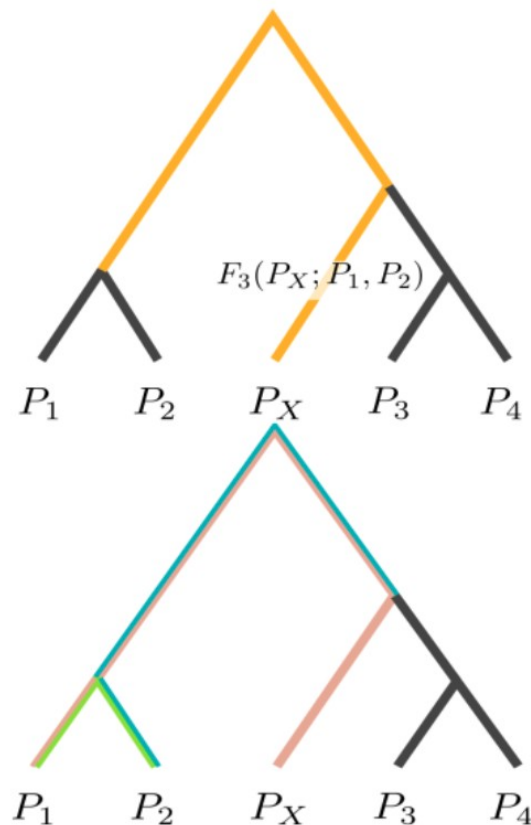
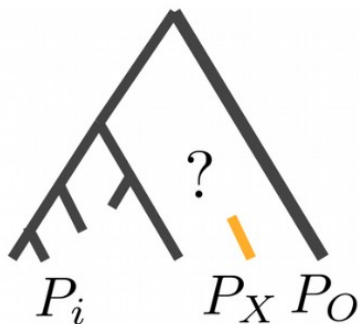
- F_2 is a covariance in contrast to F_{st} which is a correlation coefficient, therefore F_2 is tree-additive by its definition

$$F_2 = 2\pi_{12} - \pi_1 - \pi_2$$

- F_3 can be expressed through a combination of F_2 -statistic

$$F_3(P_X; P_1, P_2) = \frac{1}{2}(F_2(P_X, P_1) + F_2(P_X, P_2) - F_2(P_1, P_2))$$

- Outgroup F_3 -statistic: for an unknown population P_X find the closest population from a panel P_i computing the distances via an outgroup P_O .



David Reich Lab
Ancient DNA, Biology, and Disease

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Each dataset consists of four files, in [eigenstrat](#) format. For details, please see: [eigensoft](#):

anno: Rich meta-information for each individual

And : Three columns: Individual ID, sex determination, and group label (population).

snps : Information on each analyzed SNP position (SNP id, physical/genetic location and reference/variant alleles, where the reference allele matches hg19).

geno: Genotypes (see note 2 below)

Version v54.1.pl

Description	anno	inc	snp	geno	Tarball all files	Notes
1240k	link (1.6 Mb)	link (666 Kb)	link (75 Mb)	link (4.8 Gb)	link (4.8 Gb)	16399 unique individuals 99990 ancient, 6399 present-day ¹
1240K+HO	link (7.1 Mb)	link (718 Kb)	link (36 Mb)	link (2.9 Gb)	link (2.9 Gb)	20503 unique individuals 99990 ancient, 10513 present-day ¹

1: includes one ancestral reference, and three present-day references: human, chimp, gorilla.

2: genotypes are in a binary form using the 'packedancestrymap' format described in in [eigensoft](#); this may be converted to a (large) text file (i.e. 'eigenstrat' format, using the software 'convert')

3) *md5sums* are available: [here](#). These may be used to verify that files which are downloaded match this distribution, using the linux command: `'md5sum <file>'`

Please note: The unique individual identifier is given in the 'Master ID' field. Multiple representatives of the same individual are thus indicated by a duplicated master ID. Some individuals are represented more than once to reflect different versions of processing or different publications. This may happen for example, when increased coverage has been generated after an initial publication. For many analyses it may be necessary to select only one version: for example the single sample L0schbour (master ID=I0001) is represented by two Version IDs, L0schbour_SNPAD.DG and L0schbour_published.DG. It would be incorrect to consider these as two samples from the same population. If it is not important which version is used, we suggest choosing the master ID which has the highest number of SNPs hit on autosomal targets.

In addition, mitogenomes for samples generated at the Reich Lab are available [here](#).

We would be grateful if users of this dataset could alert us to any errors they detect and help us to fill in missing data. This could include: (1) errors or missing information for location, latitude, longitude, archaeological context, date, and group label, (2) concerns about Y chromosome or mitochondrial DNA haplogroup determinations, and (3) evidence for other problems in the data or annotations for individuals. Please write to Swapan_Shop_Mallik and [David Reich](mailto:David_Reich) with any suggestions. We would also be grateful if members of the community could suggest additional content that would be helpful to add to this page to make it maximally useful. Finally, please let us know if there are any published ancient or modern DNA datasets that are not included in this compendium and that should be. The following datasets are already processed for inclusion in the next release which will be v55. However, if you write to us to let us know of any missing datasets not on this list we will make a special effort to include them.

```
1 head data/DavidReich/AADR.ind
```

ALT-116 M	Tubalar
ALT-117 M	Tubalar
ALT-165 M	Tubalar
ALT-845 M	Tubalar
ALT-846 M	Tubalar
GE0-002 M	Georgian
GE0-005 M	Georgian
GE0-010 M	Georgian
GE0-015 M	Georgian
GE0-020 M	Georgian

```
1 head -n 8 data/DavidReich/AADR.snp
```

rs3094315	1	0.020130	752566 G A
rs7419119	1	0.022518	842013 T G
rs13302957	1	0.024116	891021 G A
rs6696609	1	0.024457	903426 C T
rs8997	1	0.025727	949654 A G
rs9442372	1	0.026288	1018704 A G
rs147606383	1	0.026665	1045331 G A
rs4970405	1	0.026674	1048955 A G

```
1 head data/DavidReich/AADR.geno
```

[illegible]

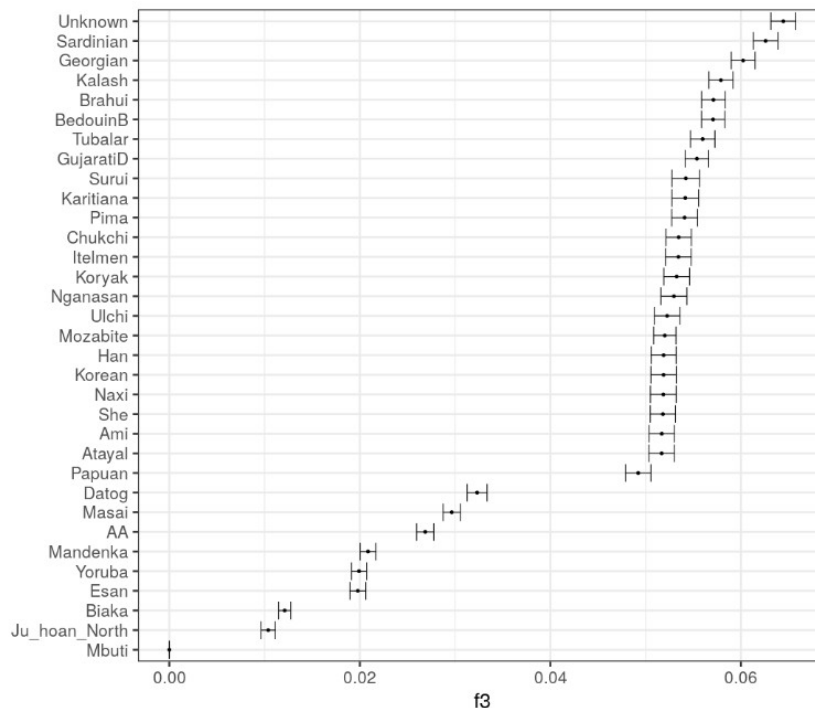
<https://reich.hms.harvard.edu/allen-ancient-dna-resource-aadr-downloadable-genotypes-present-day-and-ancient-dna-data>

Example of F3-outgroup-statistic for an Unknown population

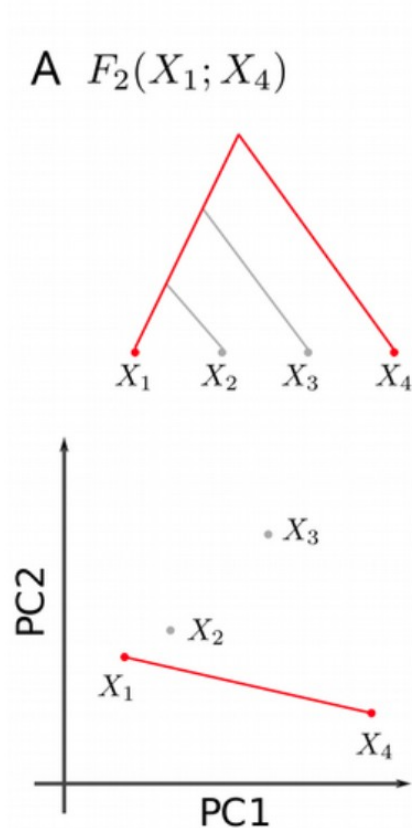
We will use the AADR dataset and compute F3-statistic for a (presumably!) European population, which is marked as **Unknown** in the dataset, with respect to a panel of other World populations using a Mbuti as an outgroup population.

```
1 library("admixtools")
2 library("tidyverse")
3
4 admixtools::extract_f2("AADR", outdir = "f2_AADR")
5
6 f2_aadr <- read_f2("f2_AADR")
7 fst_aadr <- fst(f2_aadr)
8 fst_aadr
9
10 f3_aadr <- f3(f2_aadr, pop1 = "Mbuti", pop2 = "Unknown", pop3 = unique(ind$pop)) %>%
11   arrange(est)
12 f3_aadr
13
14
15 viz_outgroup_f3 <- function(f3_res) {
16   f3_res %>%
17     mutate(pop3 = fct_reorder(pop3, est)) %>%
18     ggplot(aes(x = pop3, y = est, ymin = est - 3 * se, ymax = est + 3 * se)) +
19     geom_point() + geom_errorbar() + coord_flip() + theme_bw(25) + xlab(NULL) +
20     ylab("f3")
21 }
22 viz_outgroup_f3(f3_aadr)
```

Can you guess what was the Unknown population?



Relation between F-statistics and PCA



(Peter, 2022)

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Research



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One contribution of 15 to a theme issue 'Celebrating 50 years since Lewontin's

A geometric relationship of F_2 , F_3 and F_4 -statistics with principal component analysis

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Principal component analysis (PCA) and F -statistics *sensu* Patterson are two of the most widely used population genetic tools to study human genetic variation. Here, I derive explicit connections between the two approaches and show that these two methods are closely related. F -statistics have a simple geometrical interpretation in the context of PCA, and orthogonal projections are a key concept to establish this link. I show that for any pair of populations, any population that is admixed as determined by an F_3 -statistic will lie inside a circle on a PCA plot. Furthermore, the F_4 -statistic is closely related to an angle measurement, and will be zero if the differences between pairs of populations intersect at a right angle in PCA space. I illustrate my results on two examples, one of Western Eurasian, and one of global human diversity. In both examples, I find that the first few PCs are sufficient to approximate most F -statistics, and that PCA plots are effective at predict

